Exercise 7

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```r
# Data and Parameters
n <- 200
p <- 2
r_hat <- c(0.13, 0.13, 0.12)
lags <- 1:3

# 1. Individual Evaluation
# Standard error for white noise (approximate)
se_white <- 1/sqrt(n)
z_scores <- r_hat / se_white

# 2. Joint Evaluation (Ljung-Box Q-statistic)
# Q = n(n+2) * sum(rk^2 / (n-k))
Q_stat <- n * (n + 2) * sum(r_hat^2 / (n - lags))

# Degrees of freedom for AR(p) is H - p
df_val <- length(r_hat) - p
p_value <- 1 - pchisq(Q_stat, df = df_val)

# Output Results
cat("Individual Z-scores:", z_scores, "\n")
```

```
## Individual Z-scores: 1.838478 1.838478 1.697056
```

```r
cat("Ljung-Box Q-stat:   ", Q_stat, "\n")
```

```
## Ljung-Box Q-stat:    9.832334
```

```r
cat("Degrees of Freedom: ", df_val, "\n")
```

```
## Degrees of Freedom:  1
```

```r
cat("P-value:             ", p_value, "\n")
```

```
## P-value:             0.001714706
```

Individually:

Individually, none of the residual autocorrelations exceed the standard 5% significance threshold of 1.96 standard errors ($0.13 < 1.96 * 0.0707$). While they are all positive and relatively large, they do not technically fail an individual two-tailed test.

Jointly

The AR(2) specification is rejected when looking at the lags jointly.The Q-statistic of 9.832334 is much larger than the χ^2 critical value for 1 degree of freedom 3.84.The resulting p-value 0.0017 is well below 0.05.